Lander Zulueta

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Summary

Physics graduate (ranked 1st in my cohort) and full-time experimental researcher in quantum sensing at Qubiz.team, building and operating a fiber-based NV-ensemble magnetometer (CW-ODMR, pulsed Rabi, T_1). Hands-on experience in microwave/RF engineering (waveguide design in HFSS and COMSOL, bench-top NMR), cryogenics and low-temperature STM at CIC nanoGUNE. Complemented by numerical modeling of NV dynamics (seven-level model, Lindblad equation) and a thesis in quantum metrology. Seeking an experimental research internship or PhD in NV-based quantum sensing.

Education

B.Sc. in Physics 2022 – 2026
University of the Basque Country (EHU) *Leioa, Spain*

- Grade: 9.06/10, ranked 1st in the Physics degree.
- Coursework: Quantum Mechanics, Thermodynamics, Statistical Mechanics, Electromagnetism, Solid State Physics, Advanced Laboratory Techniques.
- **Thesis** (10/10, Honors): *Critical Metrology* (see Research Experience).

Research Experience

Full-Time Researcher, Quantum Sensing Sep. 2026 – present
Qubiz.team *Bilbao, Spain*

- Built and operate a fiber-based NV-ensemble setup (CVD diamond) for magnetic field sensing.
- Measured CW-ODMR ($\sim$1.5% contrast), pulsed Rabi oscillations and T_1 relaxation.
- Designed and simulated microwave waveguides for hyperpolarization setups with Ansys HFSS and COMSOL RF.

Research Intern, Quantum Probe Microscopy Group Jun. – Aug. 2026
CIC nanoGUNE (Supervisor: Prof. Nacho Pascual) *San Sebastián, Spain*

- Operated a low-temperature STM (LT-STM), including LHe/LN_2 transfers and sub-Kelvin cryostats.
- Characterized the transmission function of a cryogenic microwave antenna integrated into the STM junction, laying the RF calibration groundwork for ESR-STM measurements.
- Prepared STM tips and Au(111) substrates (UHV annealing, sputtering, MBE) and synthesized graphene nanoribbons by on-surface synthesis, verified by STM topography.

Researcher, Quantum Sensing Jun. – Dec. 2025
Qubiz.team *Bilbao, Spain*

- Contributed to early-stage projects in biomedicine, automotive and electrical industries, studying quantum sensing devices with a focus on NV centers.
- Built RF components of a bench-top NMR device and tested classical machine-learning methods to enhance its signal.

Undergraduate Researcher (Bachelor's Thesis) Sep. 2025 – Jul. 2026
EHU (Supervisors: Iñigo Egusquiza, Jagoba Apellaniz; Ikasiker grant) *Leioa, Spain*

- *Critical Metrology.* Question: can quantum metrology gain from operating near a quantum phase transition? Method: used the superradiant transition of the quantum Rabi model in the classical-oscillator limit. Result: showed that the precision of parameter estimation diverges at the critical point in the static limit.
- Analytic work with Schrieffer-Wolff and bosonic Bogoliubov transformations; numerics with dense diagonalization and numerical differentiation.
- Code: github.com/Landerzl/Critical-Metrology

Selected Projects

- **Computational analysis of NV centers.** Used the seven-level model and Lindblad's equation to reproduce known NV behavior (GSLAC, ESLAC, CW-ODMR, Ramsey interferometry). Used PySCF (Hartree-Fock and DFT) to study the effect of an external electric field on zero-field splitting. Code: github.com/Landerzl/NV-Centers-Code.
- **BB84 QKD experiment.** Built an experiment at EHU to exchange messages with the BB84 protocol, including optical alignment and operation of the hardware.

Honors and Awards

- **Extraordinary Degree Award in Physics**, EHU (2026): best academic record of the 2025–26 cohort.
- **Ikasiker grant**, Basque Government (2026): competitive grant for the development of the Bachelor's Thesis.
- **Honors in 15 subjects** (2022–2026): Bachelor's Thesis (Critical Metrology), Solid State Physics I & II, Quantum Physics, Nuclear and Particle Physics, Experimental Techniques II, III & IV, Sensors and Actuators, Electromagnetism II, Modern Physics, Vector and Complex Analysis, Algebra I, Chemistry II, Fundamentals of Programming.
- **Academic Acceleration** (2021): skipped one year of secondary school for outstanding academic performance.

Courses and Certifications

- Quantum Machine Learning and Quantum Optimization (36h), University of Deusto, Nov. 2026 (upcoming)
- Laser Safety Officer (16h), ProCareLight, Sep. 2026
- Quantum Chemistry with VQE (5h), IBM, Sep. 2026
- Introduction to Ansys HFSS (2h), Ansys, Sep. 2026
- Quantum Communication and Quantum Networks (4h), TU Braunschweig, Feb. 2026
- General Formulation of Quantum Information (10h), IBM, Nov. 2025
- Fundamentals of Quantum Algorithms (10h), IBM, Aug. 2025
- Quantum Sensing and Imaging (30h), Qureca, Jul. 2025

Seminars and Workshops

- Hands on Quantum Information (3 ECTS), EHU Quantum Center, Mar. 2026 – present
- 8th Basque Quantum Science and Technology Workshop / 4th BasQ/IKUR Quantum Science and Technology Workshop, Oct. 2025
- Quantum Computing: from Fundamentals to Applications (1 ECTS), BasQ, DIPC, Sep. 2025
- Quantum Technologies: Present and Future, Nquire, Jun. 2025

Technical Skills

Experimental: Fiber-based NV ensemble sensing (CW-ODMR, Rabi, T_1), microwave/RF delivery, cryogenics, UHV and low-temperature STM
Programming: Python (QuTiP, scqubits, PySCF, NumPy, SciPy), MATLAB, Wolfram Mathematica, Fortran 90
Simulation and RF: COMSOL Multiphysics, Ansys HFSS, Qorvo MatchCalc, Smith charts, LAMMPS, Fusion 360, PSpice
Tools: LaTeX, Git/GitHub, LabVIEW, Linux (WSL)
Languages: Basque (native), Spanish (native), English (C1, Cambridge, Sep. 2025), French (basic)

References

Prof. Nacho Pascual, CIC nanoGUNE (internship supervisor)
Prof. Iñigo Egusquiza, University of the Basque Country (EHU) (thesis supervisor)
Prof. Jorge Casanova, University of the Basque Country (EHU) (co-founder of Qubiz.team)